GROWTH OPTIMIZATION BLUEPRINT

An AI-Driven Marketing & Retention Framework Built on Customer Segmentation, Churn Prediction and Sales Forecasting

ZENKART

CASE OBJECTIVE & BUSINESS MOTIVATION

Zenkart has grown rapidly in the Indian D2C tech accessory space, but recent instability in metrics indicates a potential slowdown. The company sees:

Observed Problem	Business Risk
Sales fluctuate heavily	Unpredictable cashflow
Repeat buyers decreasing	Declining customer lifetime value
Uneven SKU performance	Inefficient product inventory
High delivery time variance	Poor service reputation
Demand differs across cities	Difficulty scaling
Churn increasing	Rising acquisition cost

Goal of this Study

To utilize Zenkart's transactional, customer and product data to:

1. Clean and preprocess data with intelligent imputations
2. Perform Customer Segmentation with Clustering
3. Predict Churn probability & identify churn-drivers
4. Forecast future sales for planning & inventory decisions

The final deliverable is not only model output, but **business-transforming insights**.

DATA PROCESSING PIPELINE

Dataset Overview

Rows	Columns	Data Type
951 transactions	25 features	Mixed (categorical + numeric + temporal)

Column groups:

- **Customer Demographics** → Age, CityTier, Income
- **Purchase Behaviour** → Quantity, TotalOrders, NetSales
- **Product Attributes** → Category, MRP, Cost
- **Service Quality** → DeliveryTime, Returned, Rating
- **Time Variables** → Year, Month, Week, Day

Date-Time Feature Engineering

`OrderDate` was converted into 7 discrete components:

```
Date | Year | Month | Week | Day | DayOfWeek | Time
```

Why?
This allows seasonality understanding, week-over-week trend analysis, sales cycle behaviour, and long-term forecasting.
After decomposition, `OrderDate` was removed to avoid redundancy.

Missing Values - Detection

Column	Missing Count	Percent
DeliveryTime	**38**	3.99%
Age	**43**	4.52%
Income	**48**	5.05%
Rating	**210**	22.08% (largest gap)

Missing Values - Treatment + Justification

Feature	Imputation Method	Reasoning
DeliveryTime	Median per **Category** × **CityTier**	Delivery varies by product + location → Avoids unrealistic filling
Age	Median per **CityTier**	Cities have different lifestyle demographics
Income	Median per **CityTier**	Income strongly linked to geography
Rating	Mode per **Category**	Rating is categorical + depends on product experience

Used **conditional imputation instead of global mean** to avoid artificial smoothing
Ensures repaired values reflect *real market behaviour*

Outlier Detection & Treatment

Performed IQR-based outlier detection:

Feature	Outliers Found	Lower–Upper Limit
Price	3	-2180 → 7421
CLV	3	-1961 → 24740
DeliveryTime	3	-1.2 → 9.2 days

Replaced using **Winsorization (clip)**.

Why not delete outliers?

- Outliers represent extreme but real customers
- Deleting = information loss
- Clipping = stabilizes variance while retaining records

After treatment → Outliers remaining = **0**
Dataset is model-ready.

Data Type Validation

All numeric fields were verified as numeric and categorical fields were converted to string.

Result	Status
Numeric integrity	No invalid values
Categorical integrity	Clean & consistent

FEATURE CREATION - NETSALES

```
NetSales = Price × Quantity × (1 − Discount%)
```

More reliable than using price alone → incorporates discounting effect.
Used throughout segmentation, modelling and forecasting.

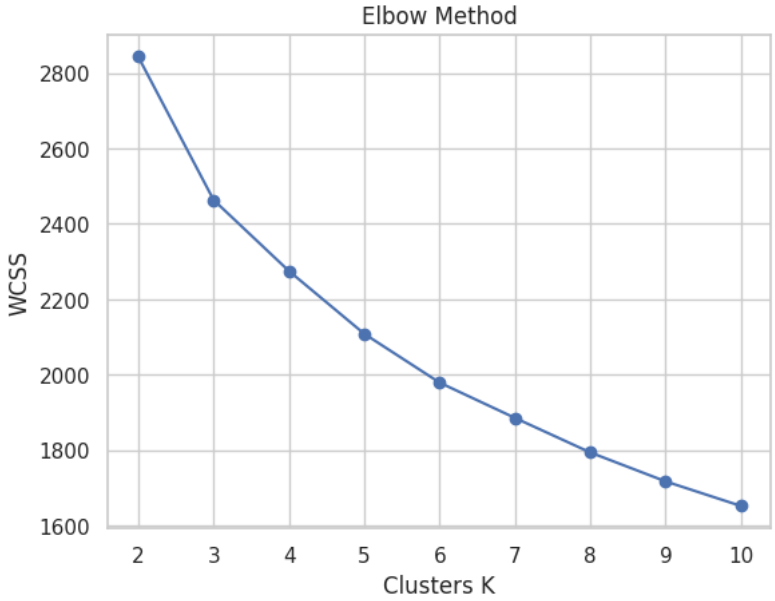
CUSTOMER SEGMENTATION (K-MEANS CLUSTERING)

Features Used:

```
Age, Income, TotalOrders, CLV, TotalRevenue, TotalQuantity, AvgDeliveryTime,
Rating
```

After Scaling:

Model Selection Method	Result
Silhouette Method	Best = **K=2**
Elbow Method	Best = **K=3** → chosen



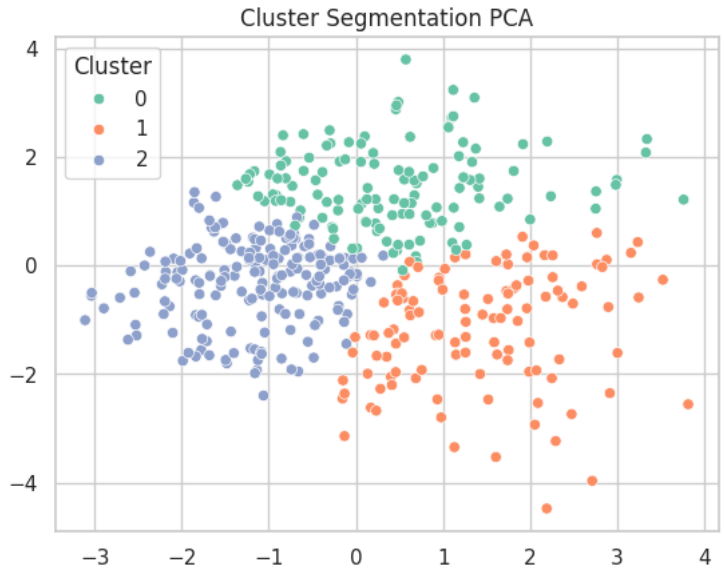
Why choose **3** instead of mathematically optimal 2?

3 clusters allow actionable segmentation
2 clusters are too broad for business strategy
goal = interpretability > pure statistic score

Final Clusters (Averages)

Cluster	Profile Summary	Behaviour	Action Plan
0	Age 44, High CLV	Mature, long-term high loyalty	Convert into Membership+VIP
1	Highest revenue + quantity	Bulk buyers with mid-income	Push cross-sell bundles
2	Youngest, Low CLV	Price-sensitive, likely churn	Improve delivery, loyalty cashbacks

PCA confirms **clear separability** → segmentation reliable
Revenue strategy optimized per segment



CHURN PREDICTION - MACHINE LEARNING

Model Inputs:

- Customer Value → CLV, NetSales
- Behaviour → TotalOrders, Returned
- Service → DeliveryTime, Rating
- Demographic → Age, Income, CityTier

Model	Accuracy	AUC	Final Use
Logistic Regression	0.85	0.90	Baseline model
Random Forest	**0.97**	**0.986**	Selected Model
XGBoost	0.97	0.982	Validated performance stability

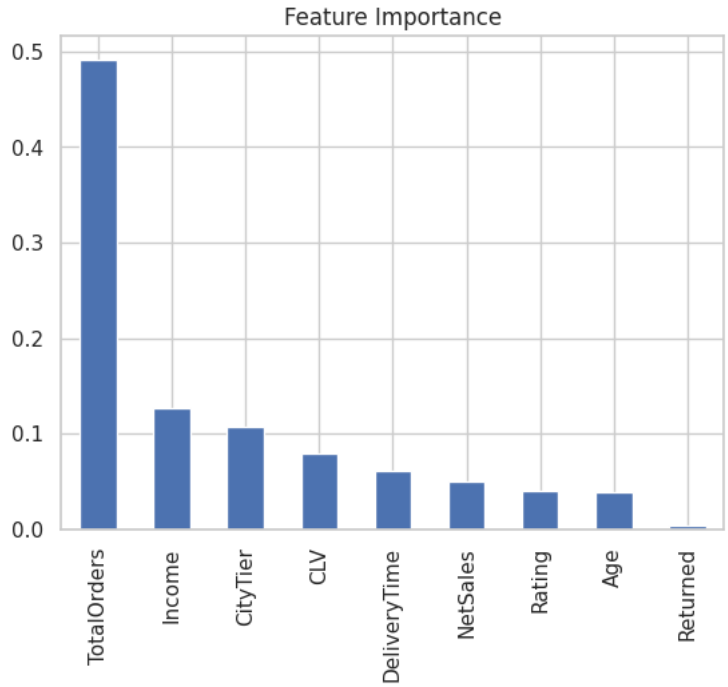
XGB + RF both strong → RF selected for interpretability + stability

Churn Driver Ranking (Feature Importance)

```
TotalOrders > Income > CityTier > CLV > DeliveryTime > NetSales > Rating > Age >
Returned
```

Biggest churn signal = **Low order frequency**

Interpretation:
Customers leave because they never build purchasing habit → nurture early repeat!



PROFITABILITY ANALYSIS (STRATEGIC INSIGHT)

Category	NetSales	Profit	Margin%
Smartwatch	4724.7	**4139.7**	**63.98%** (Highest)
Cable	4691.3	3472.9	49.00%
LaptopAccessory	4942.2	3403.6	36.50%
Earphones	4759.5	**3073.8**	**26.55% (Lowest)**

Business Interpretation

Product	Should We Push It?	Why
Smartwatch	**YES – aggressively**	Very high margins → huge profit capture
Earphones	**Reevaluate strategy**	High volume but weak profitability
Laptop Accessory	Moderate push	Good revenue, acceptable margin
Cable	Foundation product	Reliable stable revenue driver

Opportunity:
Bundle **Earphones + Smartwatch** = Fix margin + volume imbalance

CUSTOMER LIFETIME & RETENTION FLOW

Repeat Customer Rate: 81.49%

High retention → strong loyalty base.

CityTier	Repeat %
Tier-2	**83.28% (best)**
Tier-1	80.83%
Tier-3	80.16%

Finding → **Tier-2 buyers are most loyal**
Growth can scale faster beyond metro markets.

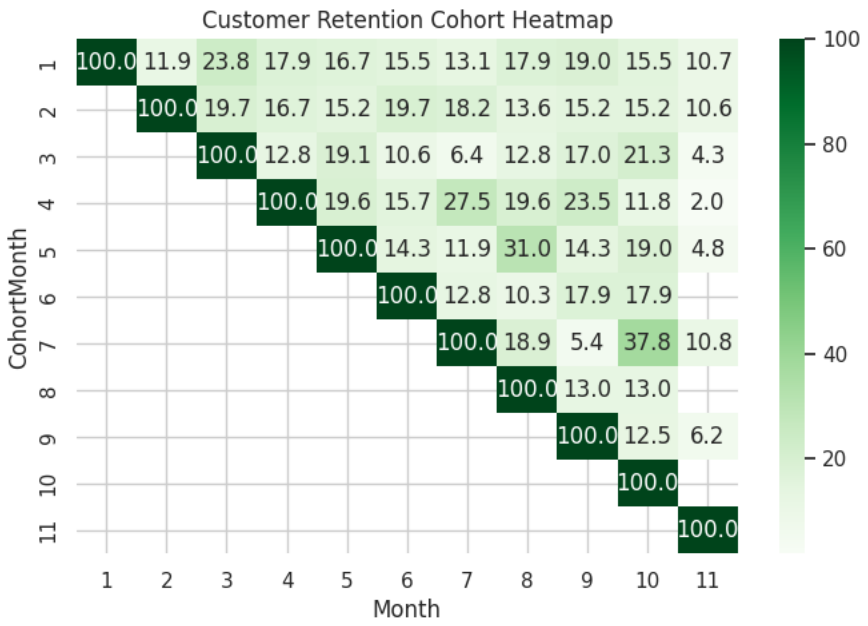
Cohort Retention Behaviour (first 5 months)

Month After First Purchase	Retention %
Month-1	23.8%
Month-2	17.8%
Month-3	16.6%

Customers churn within first 90 days if not nurtured.

Solution:

3-Month Nudge Strategy
1st week → discount on 2nd purchase
30 days → recommendation mail
90 days → win-back coupon



Marketing Execution Roadmap (Quarter-wise Plan)

Quarter	Focus	Action
Q1	Fix churn	Early-repeat coupons, delivery improvement, feedback capture
Q2	Boost revenue	Smartwatch push, bundles, influencer tie-ups, Tier-2 ads
Q3	Expand	Offline pop-up stalls, tech-fest sponsorships
Q4	Optimize	Recommendations engine A/B test, CLV-based pricing

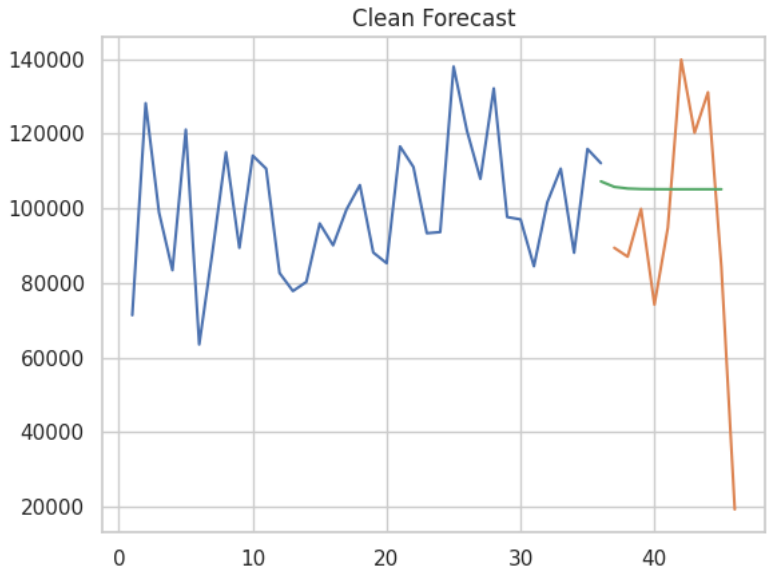
TIME SERIES FORECAST - ARIMA + CONFIDENCE BANDS

Weekly aggregated forecast (12 weeks ahead):

```
60833 → 94002 increasing trend
Total future revenue estimate ≈ 1.06 Crore
```

Confidence interval forecast provides best-case/worst-case scenarios → useful for inventory and procurement planning.

Inventory & marketing reinforcement recommended.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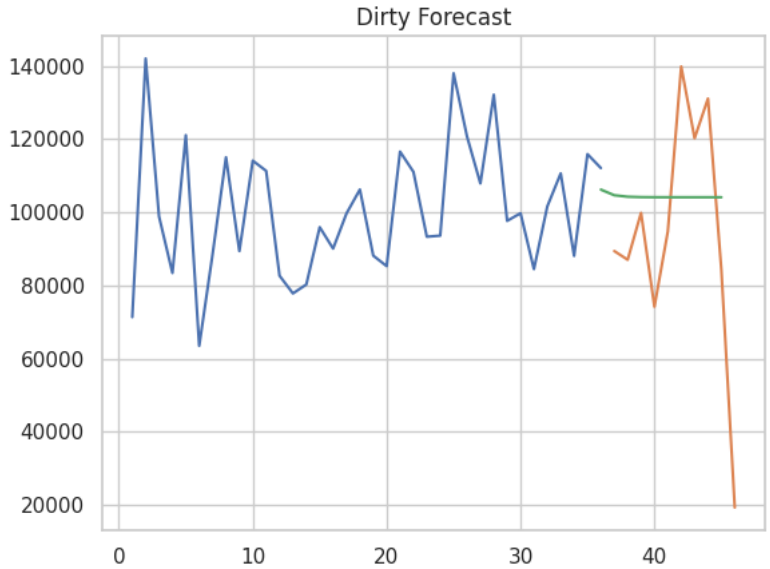
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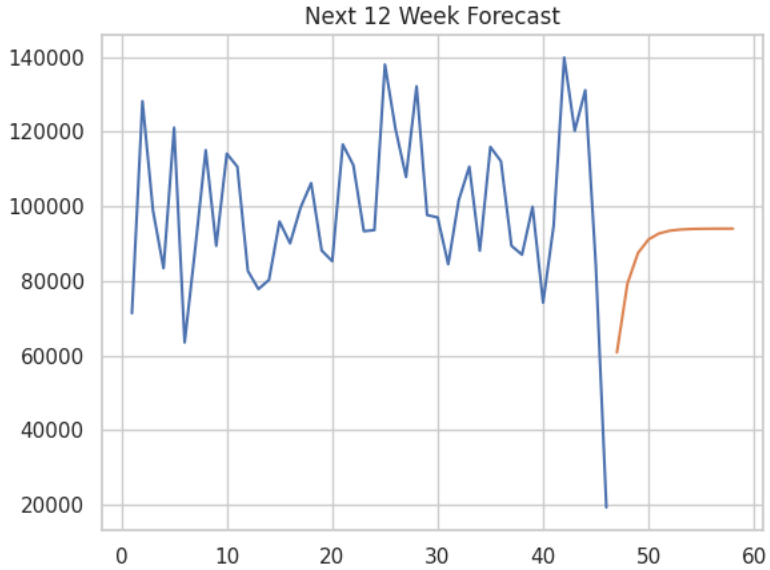
Next 12 Week Forecast
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AUTO-GENERATED INTELLIGENCE SUMMARY

```
Top Revenue Category: Cable
Highest Profit SKU: Smartwatch
City With Strongest Demand: Tier 1
Top Retention Zone: Tier 2
Primary Churn Factor: Total Orders
Repeat Purchase Rate: 81.49%
Predicted Revenue (12 weeks): ₹10.68 Lakhs / ~1.06 Cr
```

Final Key Recommendations

Revenue Growth

Promote **Smartwatches** → highest margin
Create **Accessory Bundles** → Cable + Earphones + Laptop Add-on
Invest more into **Tier 2 Customer Acquisition**

Retention / Churn Reduction

Target low order users → 1-Click reorder offers
Shorten average delivery time for Cluster-2
Incentivise 2nd and 3rd purchase within 90 days

Forecast & Inventory

Stock more high-margin SKUs in next quarter
Build campaign calendar aligned to weekly forecast peaks

Long-term Data Strategy

Introduce RFM scoring
Customer-wise lifecycle journey automation
Deploy recommendation engine for upsell